



MOTOR DATASHEET & DRAWING - SULFUR FEED PUMP

END USER: LITHIUM NEVADA LLC

PO NO.: 4550001187

TAG NO.: 1510-PP-004/5

SERIAL NO.: 77068-1/2

TYPE: VSHR-1682

PROJECT: THACKER PASS PROJECT

PROJECT NO.: HOU-23001048-A1

EQUIPMENT: SULPHUR FEED PUMP

DOC NO.: 1500-EL-VEN-4165-EXP-0102

VDRL CODE: 2701 + 2706

VENDOR DATA STATUS

- ☐ 1 : APPROVED
- ☐ 2 : APPROVED WITH COMMENTS (RESUBMIT AS FINAL)
- ☐ 3 : NOT APPROVED (REVISE AND RESUBMIT)
- ☐ 4 : APPROVAL NOT REQUIRED (FOR INFORMATION ONLY)

REVIEW DATE:

DEPT SIGNATURE:

SUBMISSION STATUS

2	11/25/2025	ISSUE FOR APPROVAL	Mason Seidel WEIR LEWIS PUMPS	Reid Chickering WEIR LEWIS PUMPS
1	10/13/2025	ISSUE FOR APPROVAL	Mason Seidel WEIR LEWIS PUMPS	Reid Chickering WEIR LEWIS PUMPS
0	4/30/2025	ISSUE FOR APPROVAL	Mason Seidel WEIR LEWIS PUMPS	Reid Chickering WEIR LEWIS PUMPS
REV	REV DATE	DESCRIPTION	PREPARED BY	REVIEWED BY

		LOW VOLTAGE (460V) INDUCTION MOTORS DATASHEET			Revision	
		Doc No. 1500-EL-DTS-3121-EXP-5001			3	

ITEM	REV	Client: Lithium Americas Corp. Project: Thacker Pass					Project No.	HOU-23001048-A1	Date	2024-08-20
1		GENERAL INFORMATION								
2		Equipment Name:		LV Induction Motor		Vendor:				
3		Number Required				Vendor Quotation Number:				
4		Equipment Number(s):				Manufacturer:		WEG (Agreement CPA 370019)		
5		Specification Number:		1999-EL-SPC-0000-BEC-0013		Model Number:				
6		Package Number:								
7		Instructions to Vendors:								
8		1. Vendors must complete Vendor Data column for each item, otherwise bid may not be considered. "VTA" denotes items for "Vendor to advise" as part of the proposal.								
9		Description		Units	Project Requirement		Vendor Information			
10		SITE CONDITIONS								
11		Site Geographic, Climatic and Seismic Data			Refer to 1000-PE-SPC-0000-BEC-0001					
12		Location (Indoor / Outdoor)			Indoor / Outdoor					
13		Non-Hazardous (Y/N)			Refer to 1500-EL-ACP-0000-EXP-5201 & 1500-EL-ACP-0000-EXP-5202					
14		Area Classification			Refer to 1500-EL-ACP-0000-EXP-5201 & 1500-EL-ACP-0000-EXP-5202					
15		Division			Refer to 1500-EL-ACP-0000-EXP-5201 & 1500-EL-ACP-0000-EXP-5202					
16		Group			Refer to 1500-EL-ACP-0000-EXP-5201 & 1500-EL-ACP-0000-EXP-5202					
17		Temperature Class		°F	Refer to 1500-EL-ACP-0000-EXP-5201 & 1500-EL-ACP-0000-EXP-5202					
18		Design Life			25 years					
19										
20		ELECTRICAL SYSTEM PARAMETERS								
21		Nominal Supply Voltage		V	480V, 3 Phase, 3 Wire		480V, 3 Phase, 3 Wire			
22		Supply Frequency		Hz	60 Hz		60 Hz			
23		System Maximum Short Circuit Level		A	10 kA Symmetrical		10 kA Symmetrical			
24		Minimum Short Circuit Level for Harmonics Calculations		A	10 kA Symmetrical		10 kA Symmetrical			
25		Supply System Grounding			10 A, High Resistance Grounding		10 A, High Resistance Grounding			
26										
27		APPLICATION								
28		Enclosure Type			TEFC		TEFC			
29		Motor Type			Squirrel Cage		Squirrel Cage			
30		Drive System			VTA		Direct			
31		Operation (Fixed Speed, Variable, Multi-Speed)			VTA		N/A			
32		If Variable, Indicate Speed Range		rpm	VTA		N/A			
33		Preferred Synchronous Speed		rpm	1800		3600			
34		Full Load Speed		rpm	VTA		3650			
35		Duty			Continuous		Continuous			
36		Maximum Sound Level		dBA	Less than 85 dBA		Less than 85 dBA			
37		HP (kW) Rating (at Actual Site Elevation)		HP	VTA		50			
38		Motor Frame Size			VTA		32TSD			
39		Voltage, Phase, Frequency		V, Hz	460V, 3 PH, 60 Hz		460V, 3 PH, 60 Hz			
40										
41		INSULATION								
42	3	Service Factor			1		1			
43		Insulation Class			NEMA Class F		NEMA Class F			
44		Insulation Design Voltage		V	2000 Minimum		2000 Minimum			
45		Temp. Rise (°C)			NEMA Class B		NEMA Class B			
46		Allowable Time at Locked Rotor		sec.	VTA		20 s (hot) 36 s (cold)			
47		Permissible No. of Hot Starts per Hour			2 Minimum		2 Minimum			
48		No. of Cold Starts per Hour			3 Minimum		3 Minimum			
49										
50		STARTING REQUIREMENTS								
51		Starting Method (FV, RV, VFD)			VTA		FV			
52		If Reduced Voltage, Type of Starter Recommended			VTA		N/A			
53		If Reduced Voltage, Recommended % of Full Voltage on Starting		%	VTA		N/A			
54		Starting Current @ ½ Load, ¾ Load & Full Load		A	VTA		176.5, 264.75, 353			
55		Efficiency @ ¾ Load & Full Load		%	VTA		93, 93, 93			
56		Power Factor @ ¾ Load & Full Load			VTA		.81, .87, .89			
57		Locked Rotor Current @ 100% Rated Voltage		A	VTA		353			
58										
59		TORQUE								

		LOW VOLTAGE (460V) INDUCTION MOTORS DATASHEET		Revision	
		Doc No. 1500-EL-DTS-3121-EXP-5001		3	

ITEM	REV	Client: Lithium Americas Corp.			Project No.	HOU-23001048-A1	Date	2024-08-20
		Project: Thacker Pass						
60		Torque Design Characteristics		NEMA Design B	NEMA Design B			
61		Load Torque Requirements		VTA	Variable			
62		Full Load Torque	lb-ft	VTA	72.7			
63		Locked Rotor Torque	lb-ft	VTA	240%			
64		Allowable WK ²	lb-ft ²	VTA	Info not available			
65								
66		INSTALLATION DETAILS						
67		Operating Position (Horiz., Vert., Other)		VTA	Vertical			
68		Mounting (Foot, Sliding Base, Flange, etc.)		VTA	Flange			
69		Common Driven Equipment Baseplate		VTA	Yes			
70		Rotation (Viewed from NDE Shaft)		VTA	Both			
71								
72		BEARINGS						
73		Bearing: Type, Capacity LB (Dynamic Load Rating)		VTA	DE: 19,154 lbs NDE: 12,432 lbs			
74		Bearing Cat. No. (DE, ODE)		VTA	DE: 6312-C3 NDE: 6212-C3			
75		No. of Bearings		VTA	2			
76		Bearing Lubrication System		VTA	Grease MOBIL POLYREX EM			
77		Lubrication Interval		ABMA L10-50,000 Hours	DE: 6000h NDE: 7000h			
78								
79		ACCESSORIES						
80		Winding Temperature Detection		For Motors above 200 HP, 100 Ohm Platinum, 2/Phase	For Motors above 200 HP, 100 Ohm Platinum, 2/Phase			
81		Space Heaters		For all Motors 20 HP and above	For all Motors 20 HP and above			
82		Main Terminal Box Sized for Motor Leads & Terminations		1 Size Larger than Standard	1 Size Larger than Standard			
83		Vibration Detection, Specify		Same as for Driven Equipment Requirements	PT100			
84		Motor Ventilation Air Filters Provided		VTA	Info Not available			
85		Ground Connections, Number & Location		VTA	2 Lugs on Frame, 1 Lug in Junction Box			
86		Weight	lb	VTA	580			
87		Tests Performed		VTA, Provide List	WEG Standard			
88								
89		Note(s)						
90		1. WEG motors to be considered as per project frame agreement no. CPA 370019 Revision 0. 2. 3. 4.						
91								
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97		VENDOR COMMENTS:						
98								
99								
100								
101								
102								

DATA SHEET**Three Phase Induction Motor - Squirrel Cage**

Customer : WEIR PUMP AND VALVE SOLUTIONS, INC.

Product line : Special Motor

Frame	: 326TSD	Locked rotor time	: 16 s (hot) 28 s (cold)
Output	: 50 HP (37 kW)	Temperature rise ⁴	: 75 K
Poles	: 2	Duty cycle	: Cont.(S1)
Frequency	: 60 Hz	Ambient temperature	: -30 °C to +45 °C
Rated voltage	: 460 V	Altitude	: 1500 m.a.s.l
Rated current	: 57.4 A	Degree of protection	: IP55
L. R. Amperes	: 395 A	Cooling method	: TEFC
LRC (p.u.)	: 6.8	Mounting	: W-6
No load current	: 14.9 A	Direct of rotation ¹	: Both
Rated speed	: 3555 rpm	Noise level ²	: 76.0 dB(A)
Slip	: 1.25 %	Vibration class	: B
Rated torque	: 73.8 ft.lb	Starting method	: Direct On Line
Locked rotor torque	: 260 %	Coupling	: Direct
Pull up torque	: 175 %	Approx. weight ³	: 583 lb
Breakdown torque	: 240 %	Painting plan	: 212P
Insulation class	: F	Color	: RAL 5009
Service factor	: 1.15	Design	: B
Moment of inertia (J)	: 4.03 sq.ft.lb		

Output	Start	50%	75%	100%	Load type: -
Efficiency (%)	-	93.0	93.0	93.0	Load torque: -
Power factor	0.43	0.79	0.85	0.87	Load inertia (J=GD ² /4): -

	Drive end	Non drive end	Foundation loads
Bearing type	6312-C3	6212-C3	Maximum traction : -
Lubrication interval	4000 h	6000 h	Maximum compression : -
Lubricant amount	21 g	13 g	
Lubricant type	MOBIL POLYREX EM		

Notes:
#Thacker Pass Project

W22 NEMA

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Losses at normative operating points (speed;torque), in percentage of rated output power

P1 (0,9;1,0)	P2 (0,5;1,0)	P3 (0,25;1,0)	P4 (0,9;0,5)	P5 (0,5;0,5)	P6 (0,5;0,25)	P7 (0,25;0,25)
7.0	5.1	4.6	3.7	2.1	1.4	0.8

Standards	Specification	: MG1 - Part 10	Vibration	: MG1 - Part 7
	Tests	: MG1 - Part 12	Tolerance	: MG1 - Part 12
	Noise	: MG1 - Part 9		

This revision replaces and cancels the previous one, which must be eliminated.

- (1) When viewed from the drive end.
(2) Measured at 1m and with tolerance of +3dB(A).
(3) Approximate weight subject to changes after manufacturing process.
(4) At the rated point.

These are average values based on tests with sinusoidal power supply, subject to the tolerances stipulated in NEMA MG 1-12.

Rev.	Summary of changes		Performed	Checked	Date
Performed by	mateusbt				718962/2024
Checked by	AUTOMATICO				Page Revision
Date	23/07/2025				1/7 3

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DATA SHEET

LN10_50HP_4501474707_139268216_REV01_CERT



Three Phase Induction Motor - Squirrel Cage

Customer : WEIR PUMP AND VALVE SOLUTIONS, INC.

Product line : Special Motor

Space heater information
Voltage: 110-127 V
Output: 32-43 W

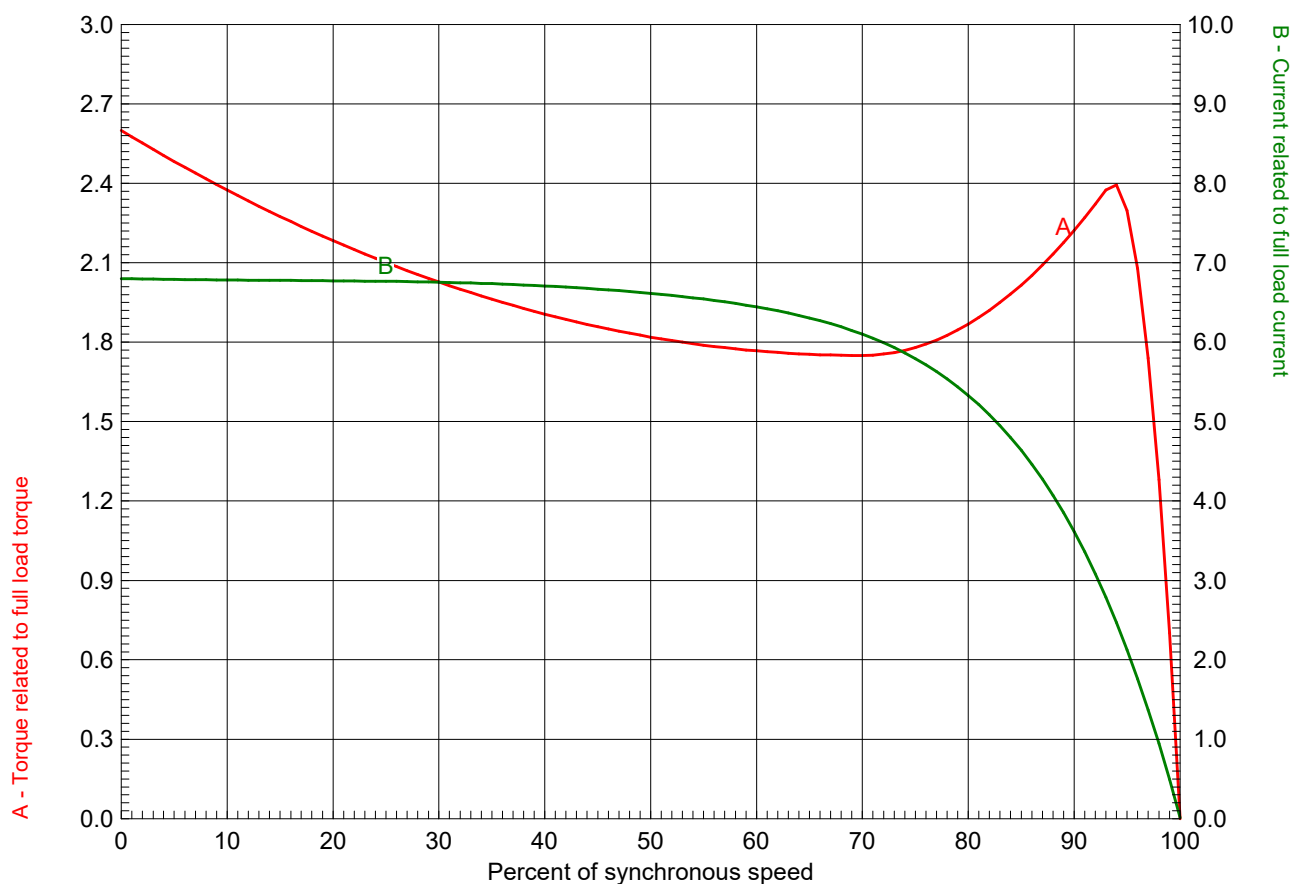
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Rev.	Summary of changes		Performed	Checked	Date
Performed by	mateusbt			718962/2024	
Checked by	AUTOMATICO			Page	Revision
Date	23/07/2025			2/7	3

TORQUE AND CURRENT X SPEED CURVE**Three Phase Induction Motor - Squirrel Cage**

Customer : WEIR PUMP AND VALVE SOLUTIONS, INC.

Product line : Special Motor

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Not subject to changes.

Performance : 50 HP (37 kW) 460 V 60 Hz 2P 326TSD

Rated current : 57.4 A
LRC (p.u.) : 6.8
Rated torque : 73.8 ft.lb
Locked rotor torque : 260 %
Breakdown torque : 240 %
Rated speed : 3555 rpm

Moment of inertia (J) : 4.03 sq.ft.lb
Duty cycle : Cont.(S1)
Insulation class : F
Service factor : 1.15
Temperature rise : 75 K
Design : B

Locked rotor time 100% : 16 s (hot) 28 s (cold)
Allowable run up time 100% : 13 s (hot) 22 s (cold)
Load inertia ($J=GD^2/4$) : 4.03 sq.ft.lb

Rev.	Summary of changes		Performed	Checked	Date
Performed by	mateusbt			718962/2024	
Checked by	AUTOMATICO			Page	Revision
Date	23/07/2025			3/7	3

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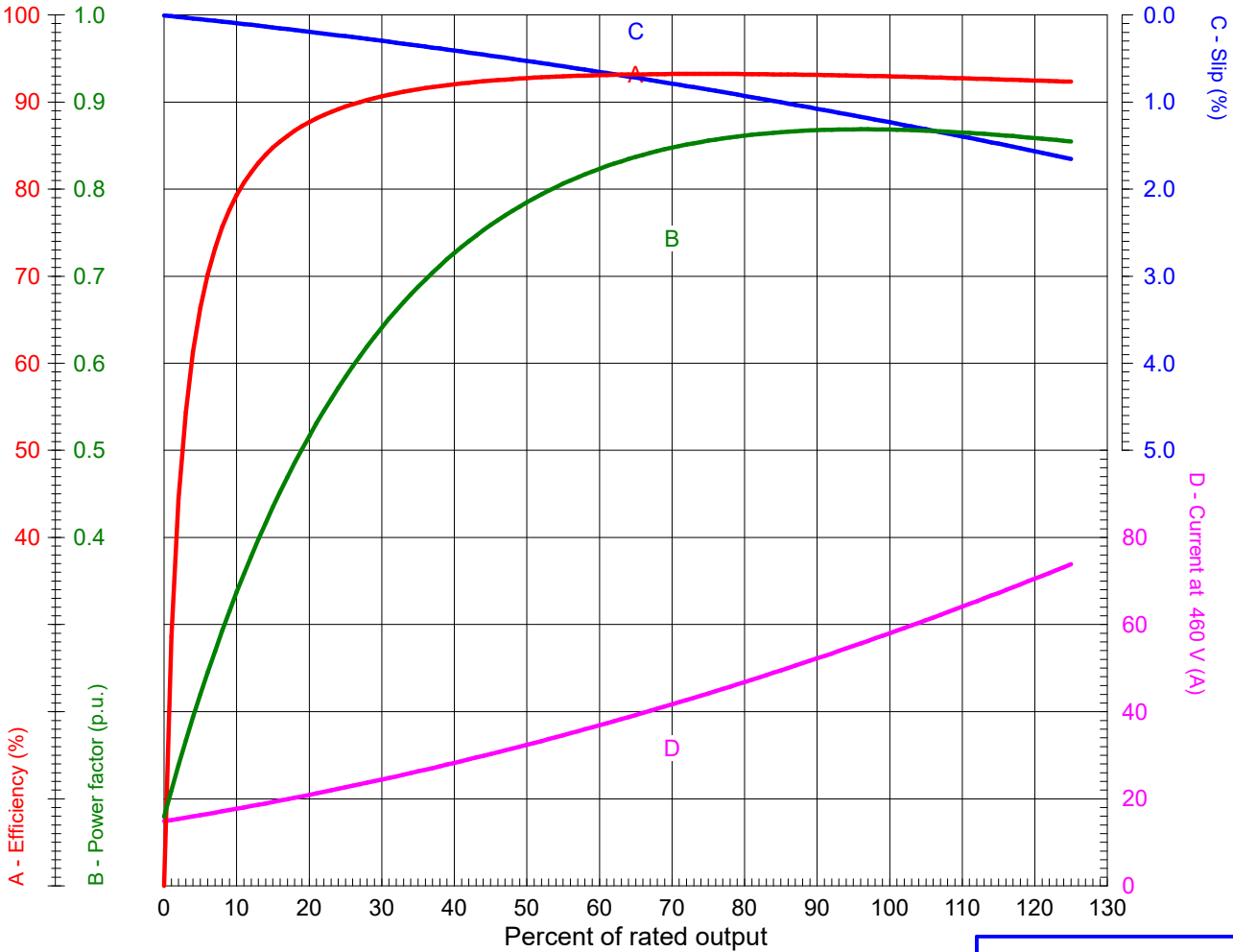
LOAD PERFORMANCE CURVE



Three Phase Induction Motor - Squirrel Cage

Customer : WEIR PUMP AND VALVE SOLUTIONS, INC.

Product line : Special Motor



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Certified document.
Not subject to changes.

Performance : 50 HP (37 kW) 460 V 60 Hz 2P 326TSD

Rated current : 57.4 A
LRC (p.u.) : 6.8
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Moment of inertia (J) : 4.03 sq.ft.lb
Duty cycle : Cont.(S1)
Insulation class : F
Service factor : 1.15
Temperature rise : 75 K
Design : B

Rev.	Summary of changes		Performed	Checked	Date
Performed by	mateusbt			718962/2024	
Checked by	AUTOMATICO			Page	Revision
Date	23/07/2025			4/7	3

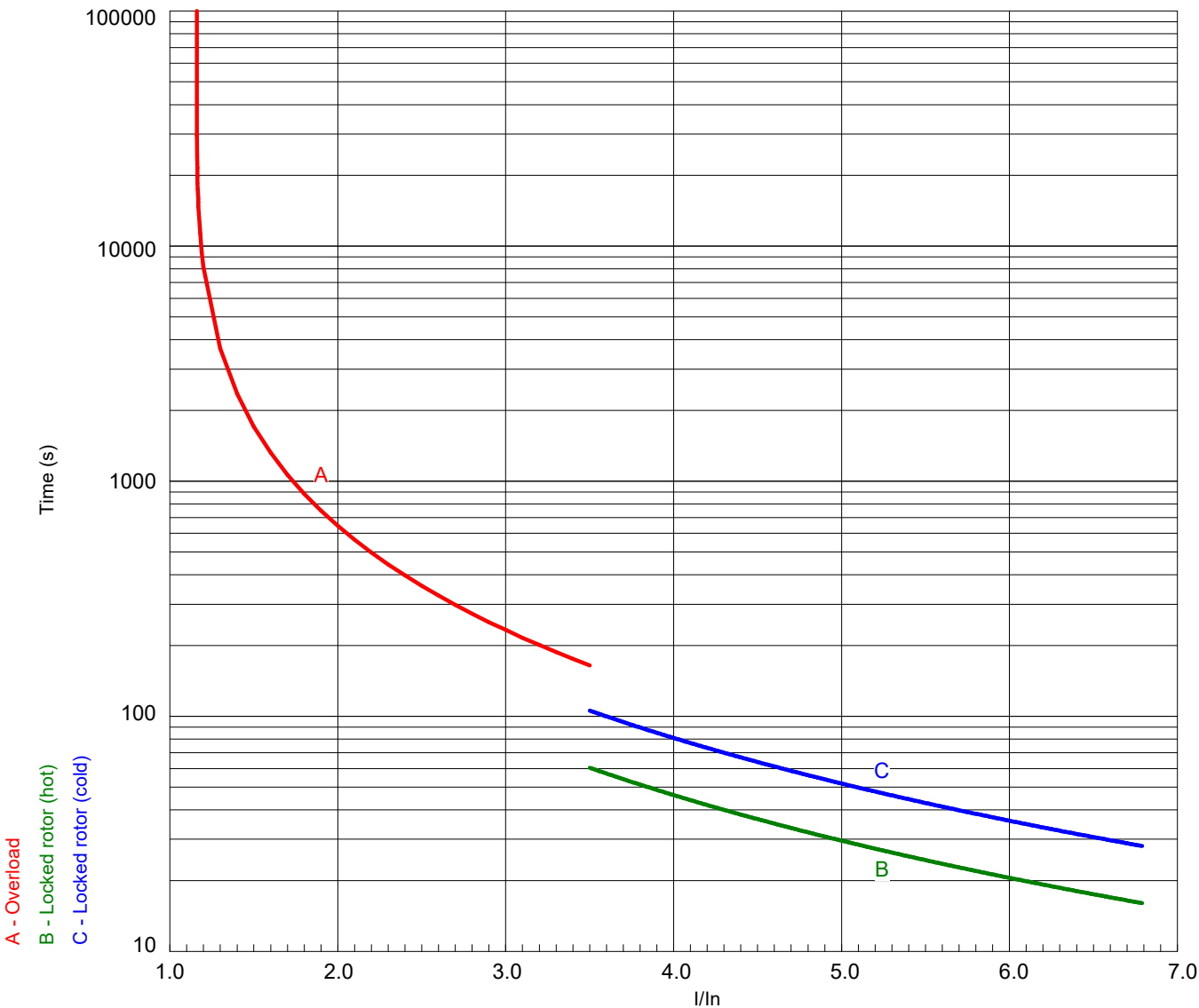
THERMAL LIMIT CURVE



Three Phase Induction Motor - Squirrel Cage

Customer : WEIR PUMP AND VALVE SOLUTIONS, INC.

Product line : Special Motor



Performance : 50 HP (37 kW) 460 V 60 Hz 2P 326TSD

Rated current	: 57.4 A	Moment of inertia (J)	: 4.03 sq.ft.lb
LRC (p.u.)	: 6.8	Duty cycle	: Cont.(S1)
Rated torque	: 73.8 ft.lb	Insulation class	: F
Locked rotor torque	: 260 %	Service factor	: 1.15
Breakdown torque	: 240 %	Temperature rise	: 75 K
Rated speed	: 3555 rpm	Design	: B

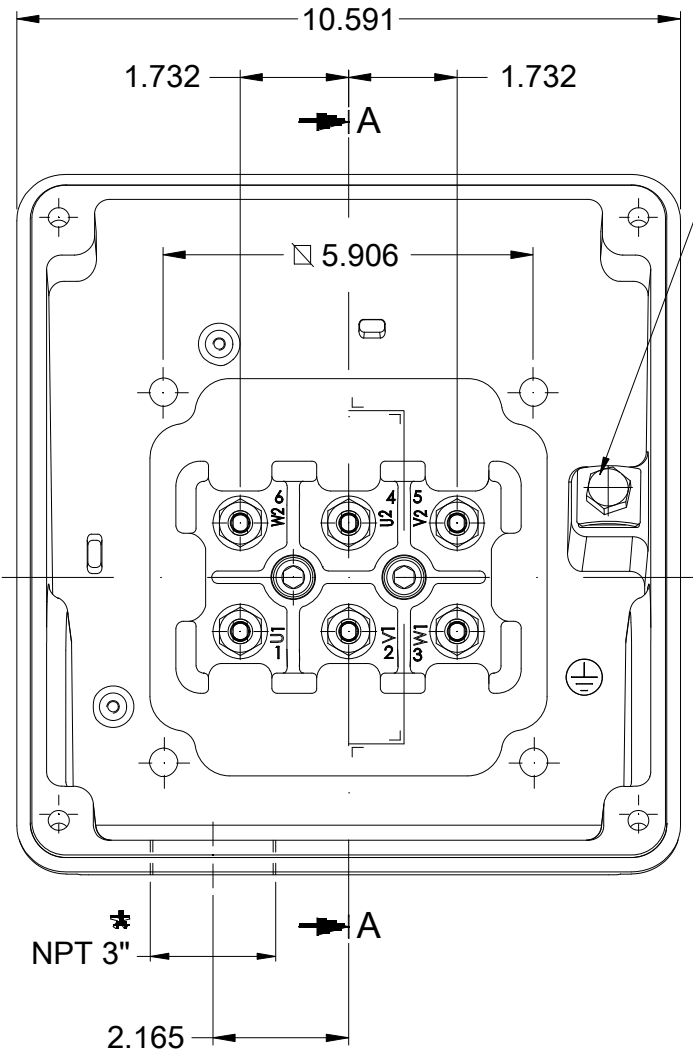
Heating constant	: 29.2 min
Cooling constant	: 87.6 min

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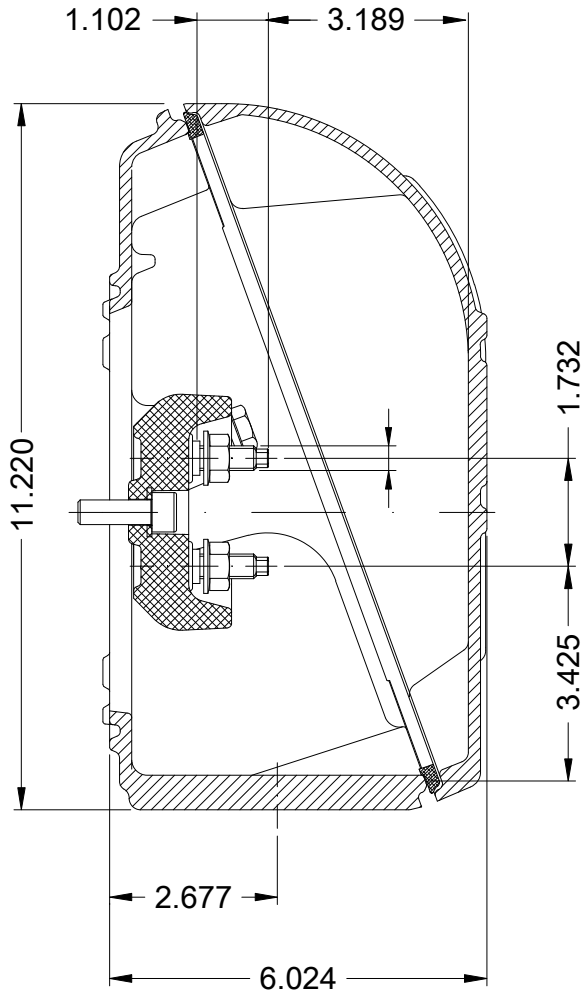
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subject to changes.

Rev.	Summary of changes		Performed	Checked	Date
Performed by	mateusbt			718962/2024	
Checked by	AUTOMATICO			Page	Revision
Date	23/07/2025			5/7	3

Dimensional drawing subject to change. We reserve the right to change constructional details.
* Detail out of scale.

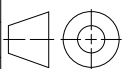


Grounding
25-50mm² / 3-1AWG

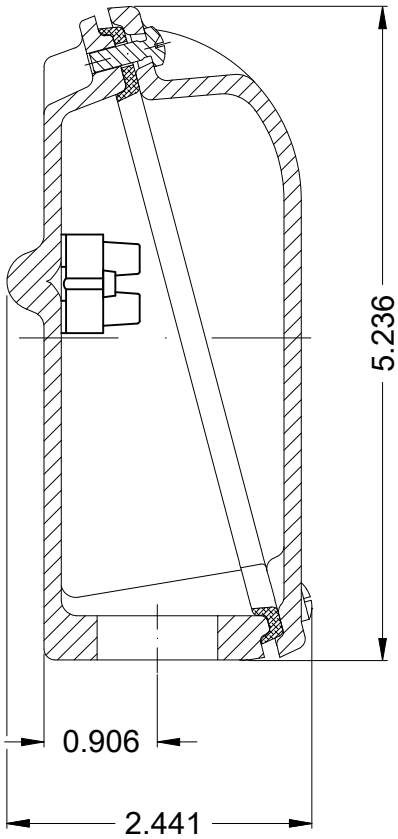
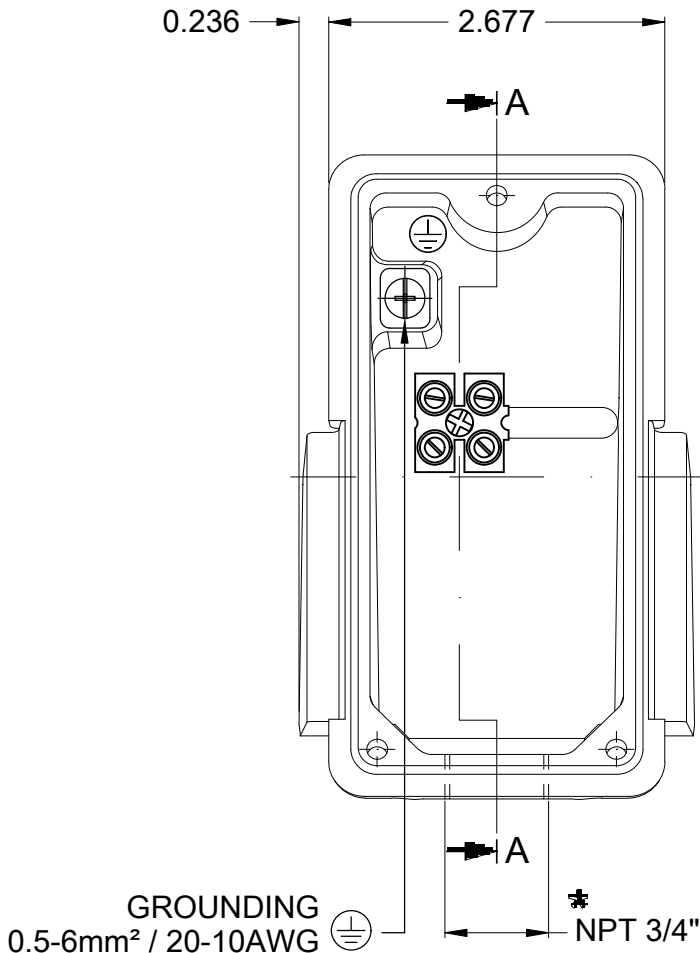


Section A-A

WEIR PUMP AND VALVE SOLUTIONS,

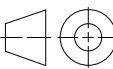
						ESC		1:3	
ECM	LOC	SUMMARY OF MODIFICATIONS				EXECUTED	CHECKED	RELEASED	DATE VER
EXECUTED	BRUNOKN		MAIN TERMINAL BOX FRAME 326T				1706209961		
CHECKED							WCD 00		
RELEASED									
REL. DATE		WEG	DULUTH			[null]	SHEET	01 / 03	

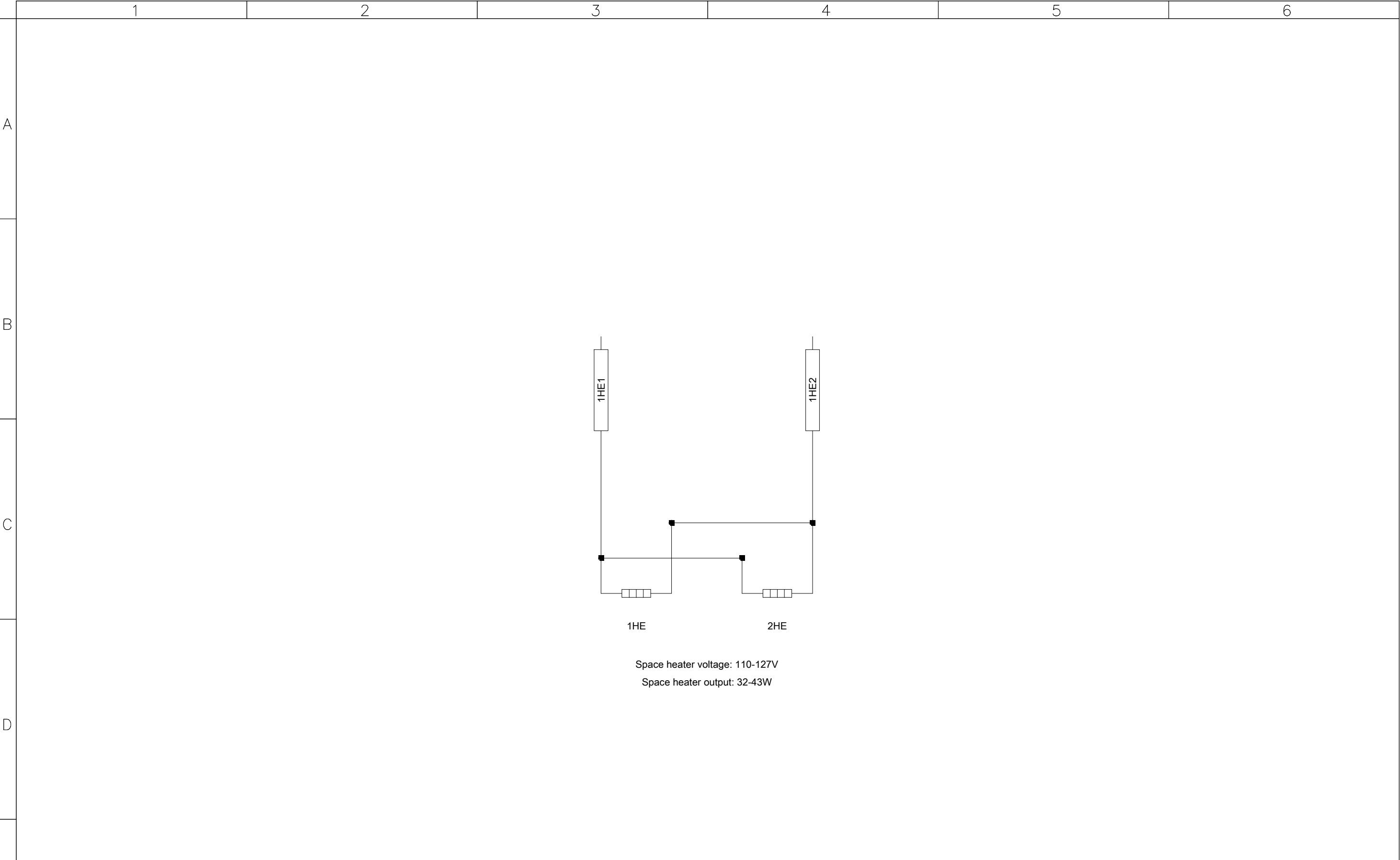
Dimensional drawing subject to change. We reserve the right to change constructional details.
* Detail out of scale.



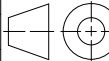
Section A-A

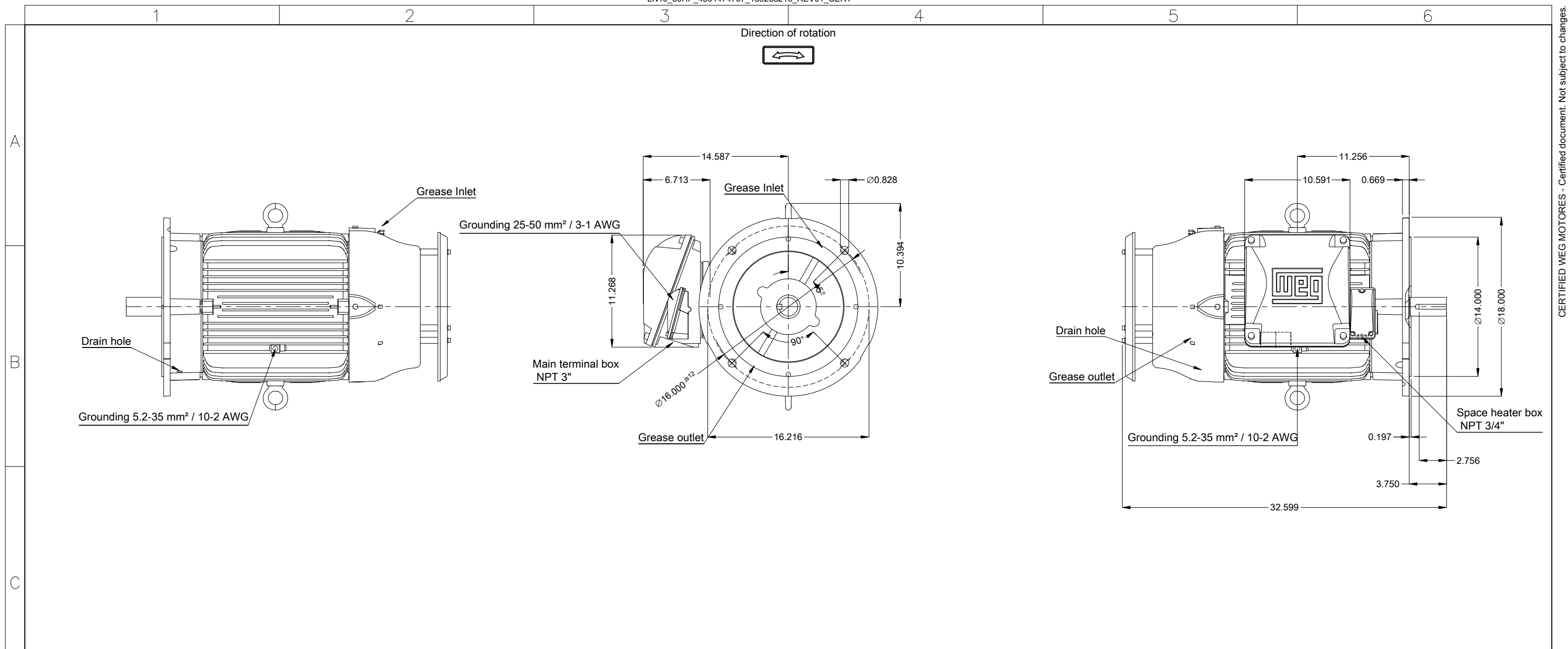
WEIR PUMP AND VALVE SOLUTIONS,

				ESC		1:1.5	
ECM	LOC	SUMMARY OF MODIFICATIONS		EXECUTED	CHECKED	RELEASED	DATE VER
EXECUTED	BRUNOKN			TERMINAL BOX FOR SPACE HEATER FRAME 326T		1706209961 WCD 00	
CHECKED							
RELEASED							
REL. DATE		WEG	DULUTH	[null]	SHEET	02 / 03	

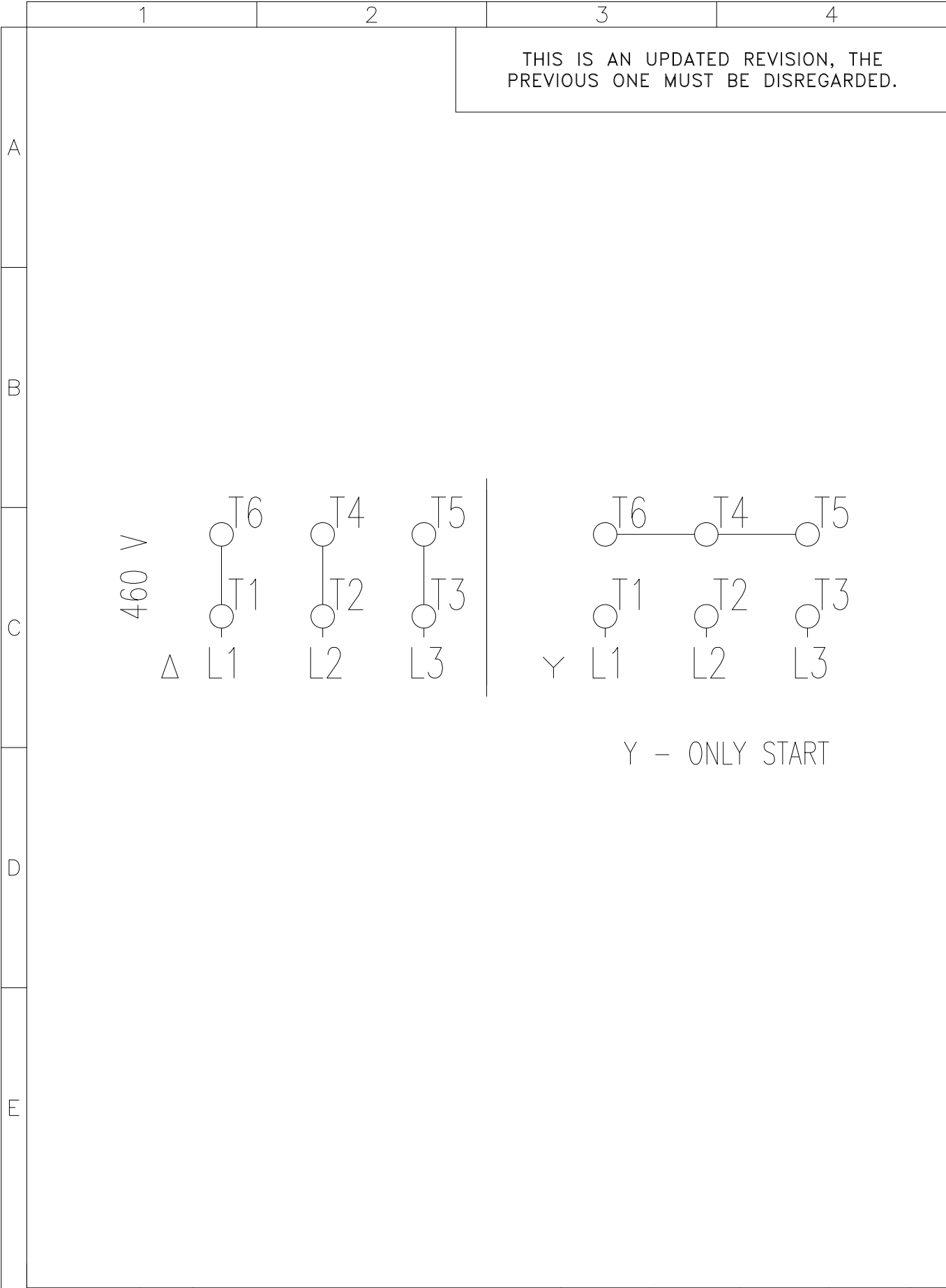


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WEIR PUMP AND VALVE SOLUTIONS,										
ECM	LOC	SUMMARY OF MODIFICATIONS				EXECUTED	CHECKED	RELEASED	DATE VER	
EXECUTED	BRUNOKN			CONNECTION DIAGRAM SPACE HEATERS			1706209961			
CHECKED										
RELEASED										
REL. DATE		WEG	DULUTH				[null]	SHEET	03 / 03	



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		ORIGINAL ISSUE			JESSICA	PATRICIA	CASTELLA	31.10.2018 00
ECM ECM	LOC LOC	RESUMO MODIFICAÇÃO SUMMARY OF MODIFICATIONS			EXECUTADO EXECUTED	VERIFICADO CHECKED	LIBERADO RELEASED	DATA DATE VER VER
EXEC. /EXECUTED	JESSICA		LOW VOLTAGE - CONNECTION DIAGRAM 460V MOTOR WITH 06 CABLES			460/6		
VERIF. /CHECKED	PATRICIA							
LIBER. /RELEASED	CASTELLA							
DATA LB /REL DT	31.10.2018	WMO	JARAGUÁ DO SUL	ADMINISTRAÇÃO DE VENDAS	FOLHA /SHEET	1 / 1		